



Sexual determination of the mandible breadth in a central Chinese population sample: a three-dimensional analysis

Mohong Deng, Rufeng Bai, Hongmei Dong, Jiao Mu, Wei Lin & Yiwu Zhou

To cite this article: Mohong Deng, Rufeng Bai, Hongmei Dong, Jiao Mu, Wei Lin & Yiwu Zhou (2016): Sexual determination of the mandible breadth in a central Chinese population sample: a three-dimensional analysis, Australian Journal of Forensic Sciences

To link to this article: <http://dx.doi.org/10.1080/00450618.2015.1134657>



Published online: 18 Feb 2016.



Submit your article to this journal [↗](#)



View related articles [↗](#)



View Crossmark data [↗](#)

Sexual determination of the mandible breadth in a central Chinese population sample: a three-dimensional analysis

Mohong Deng^{a,†}, Rufeng Bai^{b,c,†}, Hongmei Dong^d, Jiao Mu^d, Wei Lin^d and Yiwu Zhou^{d,*}

^aDepartment of Oral-Maxillary Surgery, Wuhan University Stomatology Medical College, Wuhan, People's Republic of China; ^bCollaborative Innovation Center of Judicial Civilization, Beijing, People's Republic of China; ^cChina University of Political Science and Law, Key Laboratory of Evidence Science (China University of Political Science and Law), Ministry of Education, Beijing, People's Republic of China; ^dDepartment of Forensic Medicine, Tongji Medical College of Huazhong University of Science and Technology, Wuhan, People's Republic of China

(Received 3 June 2015; accepted 14 December 2015)

In central China, there are no specialised mandible morphometric standards for sex determination. The purpose of this study was to produce a series of metrical mandible standards for accurate sex determination in the central Chinese population by CBCT (cone beam computed tomography) scans. The sample comprised 219 non-pathological maxillofacial CBCT images. Following three-dimensional reconstruction, four breadth dimension measurements were taken using SimPlant Pro 11.40. The breadth measurements were also described as the ratio. The variables were analysed using basic descriptive statistics, discriminant function analyses (DFA) and the receiver operating characteristic (ROC) analysis by SPSS 16.0. All the breadth dimensions described here were sexually dimorphic, were larger in males than in females and the differences were statistically significant. Some ratio variables were significantly different, but they were less dimorphic than the breadth dimensions per se. The sex cross-validated classification accuracies of the discriminant functions of four breadth dimensions ranged from 65.3 to 77.6% for the univariate DFA, 82.6% for the direct and stepwise multivariate DFA. The ROC analysis also illustrated the validity of four breadth dimensions for sex determination. We conclude that the mandible is a useful bone for sex determination among the central Chinese population in forensic anthropology.

Keywords: sex determination; mandibular breadth; central China; discriminant function analysis; 3D reconstruction

Introduction

Sex determination is one of the most important problems faced by forensic experts. Considering human skeletons, the sex difference mainly lies along the pelvis, skull and mandible^{1–3}. Sex determination is still a challenge in forensic work, especially when the pelvis and skull are not found. Sexual dimorphism in human mandible morphometrics has been studied extensively. Hu et al.⁴ found that the shape of the chin could be used to distinguish between males and females. The female mandible tends to have a pointed chin with a smooth and rounded gonial angle, while the male one tends to have

*Corresponding author. Email: zhouyiwu1@gmail.com

†These authors contributed equally to this work.

a square shape. In comparison with descriptive traits, metric analyses are more objective. Female mandible size was reported to average 92.4% of the male size⁵. Bigonial breadth, bicondylar breadth and ramus flexure, etc. were also showed to be valuable predictors of sex determination⁷.

As we known, sex determination is population specific, no general diagnostic criteria are currently allowed for the transfer from one population to another^{8,9}. The face, stature, and even bone features in different districts of China might be different. There are only a few osteometric studies on mandibular sex determination in the Chinese population^{10,11}. No systematic investigations have been conducted with regard to the mandible measurements in a central Chinese population until now. For this reason, we studied the mandible metric characterisation to establish its own reliable criteria of sex assessment in the central China population.

Most of the studies on mandible geometrical measurement have been conducted either directly on the wet mandibles from autopsied cadavers or on the dry bone collections. With rapidly advancing technology, it is now possible to acquire accurate skeletal measurements based on three-dimensional (3D) CT scans. One of these techniques most used in dentistry and oral-maxillofacial disease is the cone beam computed tomography (CBCT). This type of CT scanner uses a special type of technology to generate 3D images of dental structures and craniofacial bone in a single scan. Hilgers et al.¹² showed that images obtained with CBCT are allowed for precise and reliable measurement of the mandible.

In this study, we investigated the criteria that can be used to distinguish between male and female central Chinese by investigating the usual breadth characteristics of the mandible including bi-condylar breadth (BC), bi-gonial breadth (BG), bi-antegonial notch breadth (BA), and bi-mental foramina breadth (BM) on 3D views of CBCT images.

Materials and methods

Subjects

In total, 219 adult mandible CBCT images were used in this research. These were collected from the Department of Oral Radiology, Stomatology School, Wuhan University, People's Republic of China. Any sample with mandible deformation, fracture or any surgery or severe osteoporosis in the CBCT image, metabolic-endocrinological diseases in medical history was excluded. All the subjects were Chinese with Han ancestry and were born and lived in the province of Hubei (central China). Their consents to participate in this study were ensured before the study and the protocols were approved by the Medical Ethics Committee of Huazhong University of Science and Technology. The age and sex distribution of the sample is presented in more detail in Table 1. The mean ages of males and females were 40.05 (age range, 20–67) and 38.42 (age range, 23–63) years, respectively. The mean age between male and female shows no difference.

Table 1. Sex and age distribution of the utilised sample.

	<i>n</i>	Mean	SD	Minimum	Maximum
Male	111	40.05	13.67	20	67
Female	108	38.42	10.69	23	63

Methodology

For measurement, a 3D image of the mandible was reconstructed for every sample with SimPlant Pro 11.40. Four line variables of the mandible were selected in the mandible 3D image, which were measured as defined in Figure 1. These variables were commonly used in the mandible measurement and should be familiar to forensic anthropologists. The measurement was taken to the nearest hundredth of a millimetre (0.01 mm) in the 3D model. The breadth was also expressed using a ratio (see Table 2). The ratio between every two breadth measurements was calculated and six ratio variables were produced. To ensure the measurement precision, intra-observer precision was evaluated by calculating the relative technical error of measurement (rTEM) according to the literature¹³, which is less than 5% for all the measurements (see supplementary material).

Statistical analysis

The measurement data were analysed using SPSS16.0. All measurements with $P < 0.05$ were considered to be significant. Statistical descriptives such as the mean, standard deviation (SD), coefficient of variation (CV), standard error of the mean (SEM), were calculated for each variable. The normality test of the variables was conducted by the Kolmogorov-Smirnov test. The homogeneity of variance was tested using the Levene test. Differences between the mean values were determined by constructing a series of 95% confidence intervals (CI). The difference in mean values for both male and female samples was tested using independent-sample *t*-test. The most dimorphic measurements were selected into the discriminant function analyses (DFA) in order to find the optimal combination of variables that best discriminate sexes. Both univariate and multivariate DFA were applied. The ratio variables were not used for multivariate DFA in view of the independence of variables. And the multivariate DFA was conducted by direct and stepwise methods. All DFA calculated the discriminant function coefficients and the percentage of correctly sexed mandibles, respectively. All accuracies of discriminant equations were obtained by cross-validated classification. The receiver operating characteristic (ROC) analysis was also used to assess the sexual variation. Swets¹⁴ suggested

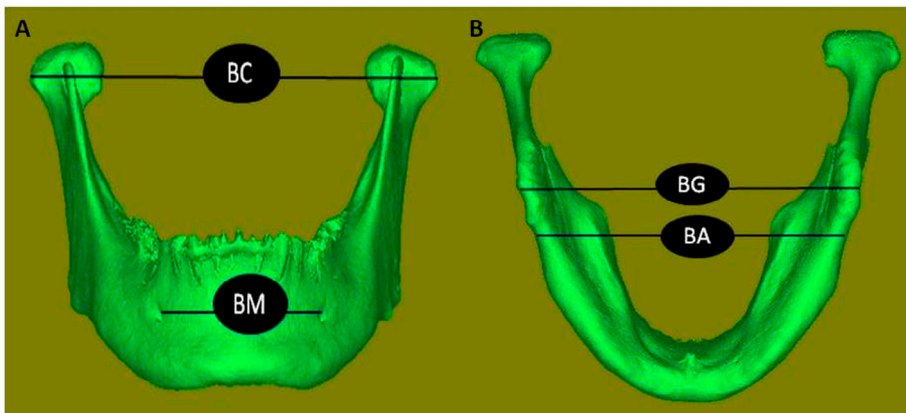


Figure 1. Landmarks selected on the 3D views of the mandibles (A) anterior view; (B) basilar view.

Table 2. A description of the measurements and the ratio on the mandible.

Variables		Description	Abb.
Measurements	Bi-condylar breadth	Direct distance between the two condylion (point 1L-1R)	BC
	Bi-gonial breadth	Direct distance between right and left gonion (point 2L-2R)	BG
	Bi-antegonial notch breadth	Direct distance between both antegonial notch depth (point 3L-3R)	BA
	Bi-mental foramina breadth	Direct distance between the two mental foramina (point 4L-4R)	BM
Ratios	BC/BG	The ratio of bi-condylar breadth to bi-gonial breadth	RCG
	BC/BA	The ratio of bi-condylar breadth to bi-antegonial notch depth	RCA
	BC/BM	The ratio of bi-condylar breadth to bi-mental foramen breadth	RCM
	BG/BA	The ratio of bi-gonial breadth to bi-antegonial notch depth	RGA
	BG/BM	The ratio of bi-gonial breadth to bi-mental foramen breadth	RGM
	BA/BM	The ratio of bi-antegonial notch depth to bi-mental foramen breadth	RAM

that the area under the ROC curve (AUC) could be used to distinguish between non-informative (AUC= 0.5), less accurate ($0.5 < \text{AUC} < 0.7$), moderate accurate ($0.7 < \text{AUC} < 0.9$), highly accurate ($0.9 < \text{AUC} < 1$), and perfect tests (AUC= 1).

Results

Variables and variable distribution

All the variables comprised of four breadth dimensions and six ratios were normally distributed (data not shown). The results for the homogeneity of variance test showed that although the probability value of RGA was lower than the threshold of significance (0.05), the other variance was generally assumed to be equal, or not dissimilar (data not shown).

Descriptive statistics and variable comparisons

The mean, SD, SEM and CV are presented in Table 3. The comparison of each variable between the sexes is reported in Table 4. All of the breadth measurements compared were significantly different ($P < 0.0001$). It was evident that all breadth dimensions of males were larger than those of females. Moreover, differences can be seen in four ratio variables ($P < 0.05$), but only RGA was significantly different ($P < 0.0001$).

Univariate DFA

The sex-dimorphic variables comprising four breadth dimensions and four ratio variables were selected for univariate DFA. The discriminant equations and cross-validated discrimination accuracies of univariate DFA are shown in Table 5. Considering the cross-validated accuracy for both sexes, BC exhibited the highest discrimination

Table 3. Descriptive statistics of the studied variables in two sexes for the measurements and ratios (males:111; females:108).

Variables	Male				Female				Female:Male
	Mean	SD	SEM	CV	Mean	SD	SEM	CV	
BC	129.70	5.02	0.48	0.04	121.80	4.56	0.44	0.04	0.939
BG	100.19	5.85	0.55	0.06	93.55	4.98	0.48	0.05	0.934
BA	89.81	4.16	0.40	0.05	85.72	4.07	0.39	0.05	0.954
BM	49.45	2.86	0.27	0.06	47.32	2.62	0.25	0.06	0.957
RCG	1.30	0.08	0.01	0.06	1.30	0.07	0.01	0.05	
RCA	1.45	0.09	0.01	0.06	1.42	0.07	0.01	0.05	
RCM	2.63	0.17	0.02	0.07	2.58	0.16	0.02	0.06	
RGA	1.12	0.06	0.01	0.05	1.09	0.04	0.00	0.04	
RGM	2.03	0.15	0.01	0.08	1.98	0.15	0.01	0.07	
RAM	1.82	0.12	0.01	0.06	1.82	0.12	0.01	0.07	

Table 4. Comparison of mean values of variables between the male and female.

Variables	Difference in means	95%CI	t-value	P-value
BC	7.898	6.619~9.176	12.173	0.000
BG	6.634	5.185~8.083	9.025	0.000
BA	4.085	2.988~5.182	7.339	0.000
BM	2.134	1.403~2.865	5.755	0.000
RCG	-0.007	-0.026~0.013	-0.697	0.486
RCA	0.024	0.003~0.045	2.250	0.025
RCM	0.050	0.005~0.094	2.215	0.028
RGA	0.025	0.011~0.038	3.654	0.000
RGM	0.049	0.009~0.089	2.430	0.016
RAM	0.004	-0.027~0.036	0.277	0.782

accuracy at 77.6%, BG had the second highest accuracy at 71. 7%, followed by BA 69.9%, and the last was BM 65.3%. Four ratio variables produced the discrimination accuracy rate, which ranged from 52.1% to 62.1%. Thus, all described breadth dimensions of the mandible can produce more than 65% or better discrimination accuracy using only univariate measurement. BC was the most useful sex-discriminating variable in the studied sample. The sex-bias varied from -1.0 of BA to 4.5 of BG in univariate DFA for four breadth dimensions (the minus sign indicating that the accuracy percentage was higher in females and vice versa).

Multivariate analyses

Table 6 shows the results of multivariate DFA. Discriminant function equations were created using direct and stepwise methods. The equations by the direct method included four breadth measurements. For the direct discriminant analyses, a cross-validated accuracy of sex classification was 82.6% with sex bias of 6.8. Only BC, BG and BM were automatically entered into the equations by the stepwise method. The result showed that the stepwise method obtained cross-validated accuracy of 82.6% with sex bias of 1.4. The cross-validated sex prediction percentage was the same by the stepwise method as by the direct method. However, the sex bias was lower in stepwise analysis than in direct analysis.

Table 5. Univariate discriminant function analysis and the correct sex classification for functions.

Function	Coefficient				Canonical correlation	Wilk's Lambda	Group centroids	Accuracy (%)			
	Standardised	Unstandardised	Eigenvalue	Original				Cross-validated			
									Male	Female	Pooled
BC	1.000	0.208	0.683	0.637	0.594	M:0.811 F:-0.834	78.4	76.9	77.6	1.5	77.6
BG	1.000	0.184	0.375	0.522	0.727	M:0.602 F:-0.618	73.9	69.4	71.7	4.5	71.7
BA	1.000	0.243	0.248	0.446	0.801	M:0.489 F:-0.503	69.4	70.4	69.9	-1.0	69.9
BM	1.000	0.364	0.153	0.364	0.868	M:0.384 F:-0.394	63.1	67.6	65.3	-4.5	65.3
RCA	1.000	12.685	0.023	0.151	0.977	M:0.150 F:-0.154	55.0	49.1	52.1	5.9	52.1
RCM	1.000	6.013	0.023	0.149	0.978	M:0.148 F:-0.152	56.8	49.1	53.0	7.7	53.0
RGA	1.000	19.994	0.062	0.241	0.942	M:0.244 F:-0.250	63.1	61.1	62.1	2.0	62.1
RGM	1.000	6.683	0.027	0.163	0.974	M:0.162 F:-0.166	57.7	56.5	57.1	1.2	57.1

Table 6. Multivariate discriminant function analysis and the correct sex classification for functions.

Function	Coefficient				Wilk's Lambda	Group centroid	Accuracy (%)			
	Standardised	Unstandardised	Eigenvalue	Canonical correlation			Original			Cross- validated
							Male	Female	Pooled	
Direct										
BC	0.723	0.151	0.884	0.685	0.531	0.923 F:-0.949	87.4	80.6	84.0	6.8 82.6
BG	0.274	0.050								
BA	0.170	0.041								
BM	0.230	0.084								
Stepwise										
BC	0.722	0.150	0.869	0.682	0.535	M:0.915 F:-0.941	84.7	83.3	84.0	1.4 82.6
BG	0.381	0.070								
BM	0.261	0.095								

ROC analysis

Results of the ROC analysis for each individual measurement are reported in Table 7. The AUC values of BC, BG, BA and BM in the study were more than 0.7. In the ratio variables, most of the AUC values were less than 0.7, except for RGA of 0.706, moreover, most were around 0.5. Among the significative AUC values, the highest mean AUC value was found among BC (0.878), the next highest mean AUC value belonged to BG (0.809), followed closely by BA (0.752), and the lowest of all breadth dimensions was BM (0.709).

Discussion

Our sample consisted of individuals in the age range from 20 to 67 because most sex-differentiating features do not become clearly visible until adulthood. Moreover, there were no differences in the age structure between two sexes, so age-related changes in the mandibles were unlikely to have influenced the results. Previous studies have reported considerable sexual dimorphism in human mandibles^{7,15–17}, but the researchers did not unanimously agree upon the relative usefulness of different mandible dimensions for sexual determination.

This study described the metric characterisation of the mandible in the central China population to define the degree of sexual differentiation. All the breadth dimensions analysed, including BC, BG, BA and BM, were significantly greater in men than in women ($p<0.0001$). Female mandible breadth size was between 93.4% and 95.7% of male size, which indicated the presence of strong sexual dimorphism in the breadth of the mandibles in the central China population sample. However, most of the ratios on the breadth dimension in the mandible are not very sex-dimorphic. The sex discrimination accuracy of their univariate DFA demonstrated these ratio variables were not as valuable as the breadth dimension per se for sex indicators. Several prior investigations also described sex dimorphism of the mandible in Chinese from different regions^{10,11–20}. The values of BC, BG and BM between spatially varying local populations are listed in Table 8. The central China sample displayed larger values of BC and BM in both sexes compared with other Chinese studies, whereas BG exhibited lower values in the central China group. Moreover, the central China group exhibited particularly strong mandibular dimorphism compared with the other regional populations. In

Table 7. Results of ROC analysis.

Variables	AUC	SE	VRP	95% CI of difference	
				Low	High
BC	0.878	0.022	0.000	0.834	0.922
BG	0.809	0.029	0.000	0.752	0.865
BA	0.752	0.033	0.000	0.688	0.816
BM	0.709	0.035	0.000	0.640	0.777
RCG	0.468	0.039	0.418	0.392	0.545
RCA	0.575	0.039	0.056	0.499	0.650
RCM	0.579	0.039	0.44	0.503	0.654
RGA	0.632	0.038	0.001	0.558	0.706
RGM	0.591	0.038	0.021	0.515	0.666
RAM	0.513	0.039	0.747	0.436	0.589

Table 8. Comparative summary statistics for the variables of the mandible in Chinese population.

Authors	Samples	Population	Variables (mean ± SD)			
			BC	BG	BM	
This study	3D image	Central Chinese	M:127.63±5.50 F:121.01 ± 4.96	M:97.95 ± 5.33 F:92.56 ± 4.43	M:50.26 ± 2.80 F:47.63 ± 2.81	$p < 0.0001$
Jin et al. ¹¹	Bone specimens	Northeast Chinese	M:122.42 F:114.42	M:103.32 F:96.72	M:46.92 F:44.10	$p < 0.05$ $p > 0.05$
Zhang et al. ¹⁰	Bone specimens	Heterogeneous Chinese		M:100.47 ± 5.53 F:95.06 ± 4.93	M:48.15 ± 3.28 F:46.09 ± 2.18	$p < 0.001$
Tan et al. ¹⁹	Cadavers	Unknown		M:101.03 ± 10.03 F:96.52 ± 8.74		$p > 0.05$
Yin et al. ²⁰	Bone specimens	Unknown	M:124.87 ± 5.74 F:116.30 ± 5.98	M:103.63 ± 5.91 F:98.17 ± 5.97	M:46.84 ± 2.31 F:46.56 ± 2.64	$p < 0.05$ $p > 0.05$
Cao et al. ¹⁸	Excavated bone	Unknown	M:125.22 ± 6.07 F:120.53 ± 5.94	M:101.60 ± 5.91 F:98.07 ± 5.97	M:46.81 ± 2.11 F:46.46 ± 2.84	$p < 0.05$ $p > 0.05$

particular, BG and BM did not exhibit sex dimorphism in some studies^{11–20}. The disagreement regarding the sex-dimorphic difference from different studies might be due to the dimorphism expression of the different population, climate conditions, dietary structure and so on²¹. Similarly, in other race groups, osteometric studies also mentioned that the breadth measurements of the mandible were very dimorphic^{2,17}. However, no significant sex-dependent difference was found in the bigonial breadth in the Egyptian population²². The current study indicates that mandible morphological distinction exists among different populations, even among Chinese in different districts. Thus, the study reaffirms the existence of population variability in mandible size and the relative expression of sexual dimorphism and the need for regional standards for discriminating sex from the mandible.

In general, the sex classification accuracy of four breadth measurements presented in the current study ranged from 65.3% (BM alone) to 77.6% (BC) by univariate DFA, while an accuracy of 82.6% was achieved for the multivariate DFA by both the direct and the stepwise methods. The allocation accuracy of multivariate DFA in the study was higher than that of univariate DFA, but the forensic application of univariate data is straightforward; furthermore, the univariate analysis can be used in different kinds of mandible remains. Either univariate or multivariate discriminant equations of the mandible had a respective advantage for sex assessment in the analysed sample. Although standards providing high-expected sex discrimination accuracy are obviously desirable, if the discrimination accuracy is associated with an unacceptably large sex-bias, then the forensic applicability of the standard will be limited. In the present study, the associated sex-bias for each function was below 7%, and even below 5% for four breadth measurements.

The accuracy obtained in the current study is comparable to that from the previously published work on other populations. When the mandible was used together with the cranium, the accuracies were higher than the current study^{2,23}. However, when the mandible was used on its own, the accuracy was close to that of our study. Steyn and Iscan²⁴ achieved an accuracy of 81.5% with five mandibular parameters (i.e. bigonial breadth, total mandibular length, bicondylar breadth, minimum ramus breadth and gonion-gnathion) in South African whites. Moreover, in Steyn's study, the bigonial breadth was demonstrated to be the most dimorphic of the measurements and even obtained average accuracies of 82%; in the current study, bigonial breadth was less dimorphic and only produced the accuracies of 71.7%. Franklin⁶ also used a stepwise analysis by selecting coronoid height, maximum mandible length, and bigonial breadth among nine variables for Indigenous South Africans and the accuracy of correct sex classification was 81.8%.

ROC analysis has become the standard tool for assessing the prediction accuracy in medical diagnosis. In forensic anthropology, ROC can be used for sex estimation through comparison of AUC value. The larger the AUC value, the better the measurement for distinguishing male from female. The ROC results in this study suggest that BC, BG, BA and BM are the sexually dimorphic dimensions. As expected, the AUC values tend to be consistent with the result of DFA.

Apart from the classical osteometric methodology, sexual determination of the skeleton was also investigated by means of radiographs and computed tomography. The CBCT technique introduces an innovation in modes of image scanning and volumetric reconstruction in dentomaxillofacial radiology. Hilgers et al.¹² found that all mandibular measurements based on CBCT views were accurate. No measurement made with CBCT differed from the anatomic truth measured directly by digital calliper.

Moreover, intraobserver CBCT measurements were highly reliable compared with anatomic truth and measurements made from the corresponding cephalogram images. To our knowledge, few studies on the mandibles by 3D CBCT images were used for sex determination. The current study is the first to be conducted on the Chinese population to evaluate sex determination using osteometric mandibular measurements by 3D CBCT images. This study showed that the sex dimorphism achieved using 3D views was similar to that reported from the direct anatomical study of Zhang in other populations¹⁰. In contrast with the direct bone measurements, 3D imaging can be used to identify semi-decomposed bodies without special preparation (e.g. maceration), and so it is more convenient. The advent of CBCT also brings the opportunity to assess small structures with a good resolution²⁵. In the current study, we chose not only the frequently used measurements such as BC, BG and BM, but also the uncommon measurement BA, which was not easily measured by the use of digital callipers. Thus, the 3D imaging technique is expected to develop more potentially valuable sex indicators in forensic anthropological practice.

Conclusion and recommendations

This study has provided a preliminary evaluation for sex assessment by some mandible osteometric standards from 3D images in central Chinese population sample. A series of forensic discriminant equations for sex estimation in this population were outlined. Our work has demonstrated that the breadth size of the mandible is a very useful indicator for sex determination in this population. More thorough investigations for seeking better sex discriminators of the mandible based on CBCT 3D images are needed. This research also suggests that virtual measurements obtained from 3D images by CBCT may serve as a substitute for direct anatomic measurements in osteological research of the maxillofacial region in forensic anthropology.

Disclosure statement

No potential conflict of interest was reported by the authors.

Funding

Funded by the National Nature Science Foundation of China (No. 81172902).

References

1. Krogman WM, Iscan MY. The human skeleton in forensic medicine. 2nd ed. Springfield, MA, 1986.
2. Giles E. Sex determination by discriminant function analysis of the mandible. *Am J Phys Anthropol.* 1964;22:129–135.
3. Scheuer L. Application of osteology to forensic medicine. *Clin Anat.* 2002;15(4):297–312.
4. Hu KS, Koh KS, Han SH, Shin KJ, Kim HJ. Sex determination using nonmetric characteristics of the mandible in Koreans*. *J Forensic Sci.* 2006;51:1376–1382.
5. Humphrey L, Dean M, Stringer C. Morphological variation in great ape and modern human mandibles. *J Anat.* 1999;195:491–513.
6. Franklin D, O'Higgins P, Oxnard CE, Dadour I. Discriminant function sexing of the mandible of indigenous South Africans. *Forensic Sci Int.* 2008;179:84. e1-. e5.
7. Loth SR, Henneberg M. Mandibular ramus flexure: a new morphologic indicator of sexual dimorphism in the human skeleton. *Am J Phys Anthropol.* 1996;99:473–485.

8. Giles E, Elliot O. Sex determination by discriminant function analysis of crania. *Am J Phys Anthropol.* 1963;21:53–68.
9. Rösing F, Graw M, Marré B, Ritz-Timme S, Rothschild M, Röttscher K, Schmeling A, Schröder I, Geserick G. Recommendations for the forensic diagnosis of sex and age from skeletons. *HOMO-J Comp Human Biol.* 2007;58:75–89.
10. Zhang JZ, Liu ZC. New method for Sex determination by analysis of the mandible. *Forensic Sci Technol.* 1996;1:33–36.
11. Jin DZ, Yu DY, Pi YH, Guo JX. Sex determination of modern human mandible by means of multiple regression method. *Yanbian university Med J.* 1985;8:195–202.
12. Hilgers ML, Scarfe WC, Scheetz JP, Farman AG. Accuracy of linear temporomandibular joint measurements with cone beam computed tomography and digital cephalometric radiography. *Am J orthod Dentofac.* 2005;128:803–811.
13. Franklin D, Flavel A, Kuliukas A, Cardini A, Marks MK, Oxnard C, O'Higgins P. Estimation of sex from sternal measurements in a Western Australian population. *Forensic Sci Int.* 2012;217:230. e1-. e5.
14. Swets JA. Measuring the accuracy of diagnostic systems. *Science.* 1988;240:1285–1293.
15. Dayal M, Spocter M, Bidmos M. An assessment of sex using the skull of black South Africans by discriminant function analysis. *HOMO-J Comp Human Biol.* 2008;59:209–221.
16. Haun SJ. Brief communication: a study of the predictive accuracy of mandibular ramus flexure as a singular morphologic indicator of sex in an archaeological sample. *Am J Phys Anthropol.* 2000;111:429–432.
17. Vodanović M, Dumančić J, Demo Ž, Mihelić D. Determination of sex by discriminant function analysis of mandibles from two Croatian archaeological sites. *Acta Stomatol Croat.* 2006;40:263–277.
18. Cao WQ, Ding SH, Liu WJ, Tian XJ. Measurement and sexual discriminant analysis of mandible. *Acta Academiae Med Qingdao.* 1998;34:117–118.
19. Tan JG, Zeng XH, Chen SH, OuYang SX, Li YB, Peng ZS, Wan W, Xiao JZ. Applied anatomic investigation of measurements of Chinese mandibular and its correlation analysis. *J Nanhua University (Medical Edition).* 2004;32:447–449.
20. Yin XZ, Zhao DM, Yang Z, Luo SB, Zhao X, Gao YB. Measurements and clinical significance of mandible. *HeiLongjiang Med Pharm.* 2012;35:85–86.
21. Frayer DW, Wolpoff MH. Sexual Dimorphism. *Annu Rev Anthropol.* 1985;14:429–473.
22. Kharoshah MAA, Almadani O, Ghaleb SS, Zaki MK, Fattah YAA. Sexual dimorphism of the mandible in a modern Egyptian population. *J Forensic Leg Med.* 2010;17:213–215.
23. Hanihara K. Sex diagnosis of Japanese skulls and scapulae by means of discriminant function. *J Anthropol Soc Nippon.* 1959;67:21–27.
24. Steyn M, İşcan MY. Sexual dimorphism in the crania and mandibles of South African whites. *Forensic Sci Int.* 1998;98:9–16.
25. Liang X, Jacobs R, Corpas L, Semal P, Lambrechts I. Chronologic and geographic variability of neurovascular structures in the human mandible. *Forensic Sci Int.* 2009;190:24–32.